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Financial Literacy, Financial Innovation, and Financial Inclusion as Mitigating Factors of the Adverse Effect of Corruption on Banking Stability Indicators

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Abstract
The purpose of this study is to examine the ability of financial literacy, financial innovation, and financial inclusion to mitigate the adverse effect of corruption on banks' credit risk, profitability, and financial stability, with this joint inclusion being the novelty explored. Furthermore, we aim to compare the results across four different groups of countries, namely African, Asian, American, and European countries. The Feasible Generalized Least Squares (FGLS) estimation results indicate that corruption increases credit risk, reducing profitability and bank stability, being these effects mitigated by financial literacy, financial innovation, and financial inclusion. Furthermore, we find that financial literacy, financial innovation, and financial inclusion reduce credit risk while increasing bank profitability and stability. These results enable policymakers and managers to promote inclusion, innovation, and financial literacy to achieve banking sector stability while combating corruption.

Keywords Financial literacy · Financial innovation · Financial inclusion · Corruption · Credit risk · Profitability · Banking stability

JEL Classification A13 · D14 · D73 · E51 · G01 · G2

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Introduction

The topic concerning the impact of corruption on bank performance is intertemporal and interesting to policymakers, bank managers, investors, and bank customers (Asteriou et al., 2021; Yunan, 2021). Certainly, this topic should interest and concern the general public, because their scarce resources are used to rescue banks to avoid systematic crises, to the detriment of needs such as health, education, and other relevant infrastructures. Individual bank crisis easily infects the entire banking sector (Albaity et al., 2019) and consequently the economy.

The phenomenon of corruption is multidisciplinary involving moral, religious, cultural, social, legal, and political aspects (Yunan, 2021). Nevertheless, corruption represents the manifestation of the self-centeredness of the agents involved. Therefore, it is a crime that can be committed by high-profile politicians, well-positioned public or private managers, or simple grassroots employees (Joseph & Smith, 2021). Corruption is a problem that spans all economies; however, it is more pronounced in underdeveloped countries, regardless of how it is calculated (Ajide, 2020; Ajide & Olayiwola, 2021; Ghardallou, 2022; Setor et al., 2021; Son et al., 2020; Trabelsi & Trabelsi, 2020). Corruption is a rather serious problem in developing countries, mainly, due to a lack of legal independence, inefficient prudential regulations, and weak internal control of banks (Asteriou et al., 2021; Ghardallou, 2022), as well as the strong state presence in the bank ownership structure.

Corruption causes negative effects on the financial performance of any organization (Van Vu et al., 2018). Therefore, it occurs when public or private agents act on behalf of the authority conferred by an institution, for the personal benefit of a restricted group (Joseph & Smith, 2021; Khan, 2021; Setor et al., 2021). Satisfying personal or narrow group interests can cause market distortions and lead to inefficient allocation of resources (Ajide & Olayiwola, 2021; Ghardallou, 2022), namely, granting bank loans to an agent because of belonging to a political group, belonging to the same religion, degree of familiarity, or friendship, without properly assessing the repayment capacity.

High levels of corruption in the banking sector act as a discriminating factor in access to credit (Asteriou et al., 2021); i.e., poor access to financial products and services drives entrepreneurs and other actors in need of financing to resort to corruption, through bribery, nepotism, and influence peddling to obtain financial resources (Hanoteau et al., 2021). Thus, Bougatef (2015), Mohammad et al. (2019), and Toader et al. (2018) report the positive effects of corruption on bank profitability and stability in their studies. These authors reason that excessive bureaucracy to obtain credit and strong risk aversion are the main incentives for using bribery in the banking sector. Therefore, in the short run, corruption allows credit expansion, being, in the long run, detrimental for banks and the overall economy (harmful) due to the increase in non-performing loans. Thus, increasing credit defaults negatively affect the stability of banks (Younsi & Nafla, 2019). Additionally, evidence of the negative effect of corruption on credit risk, profitability, and bank stability is growing (Athari & Bahreini, 2021; M. S. Ben Ali et al., 2020; Jenkins et al., 2021; Kamran et al., 2019; Mohamad & Jenkins, 2020;

Mohammad et al., 2019; Rehman et al., 2020; Toader et al., 2018; Yakubu, 2019). However, these studies converge when they highlight that the effect of corruption on banks' financial stability occurs indirectly, through credit risk, as bank officials may accept bribes in exchange for high-risk loans. But bribes end up representing extra costs of lending (Ben Ali et al., 2020).

The considerable loss of retail investors during the financial crisis has motivated policymakers and academics to consider the role of financial literacy in enhancing financial stability, as financial literacy is related to the demand for financial products and services (Grohmann et al., 2018; Klapper & Lusardi, 2020). Financial literacy is a proficiency that allows one to make informed and conscious financial decisions based on familiarity and understanding of financial products and services, specifically the associated costs, returns, and risks (Grohmann et al., 2018; Jungo et al., 2021; Klapper & Lusardi, 2020). Empirical data show that high levels of financial literacy contribute to reduced credit risk and are positively associated with banks' financial stability (Jungo et al., 2021; Klapper & Lusardi, 2020). Thus, economic agents are expected to be aware that bribery is an extra cost for loans and to be cautious about entering into any transaction. Regarding financial innovation, it enables the use of financial products and services such as savings, payments, and online credits easily and securely (Sharmila, 2019). In addition, it reduces the bureaucracy of financial institutions in customer satisfaction (Beck et al., 2015) and avoids face-to-face contact between the members of the transaction (customer and bank agent), greatly reducing the possibility of negotiation. We expect that financial innovation allows reduce the bureaucracy caused by corruption and expands the channels of using financial products and services. Financial inclusion is associated with easy access to financial products and services without any discrimination at a relatively low cost (Huang & Zhang, 2020; Ouechtati, 2022; Pham & Doan, 2020; Zhang & Posso, 2019), while corruption limits access to financial products and services. Thus, this study is aimed at investigating how financial literacy, financial innovation, and financial inclusion can reduce the adverse effect of corruption on banks' financial stability.

Previous studies by Al-Smadi (2018), Bougatef (2015), Jenkins et al. (2021), Jungo et al. (2022), Kamran et al. (2019), Mohamad and Jenkins (2020), Mohammad et al. (2019), Rehman et al. (2020), and Toader et al. (2018) analyze the effect of corruption on bank profitability and financial stability and emphasize that corruption impacts the banking sector indirectly, i.e., through non-performing loans. In general, these studies have ignored the role of literacy, innovation, and financial inclusion in the relationship between corruption and the banking sector. Authors such as Ajide (2020), Jungo et al. (2023), and Sector et al. (2021) show that financial inclusion and innovation can improve corruption control. However, these authors did not analyze the effect of financial literacy on corruption, nor did they perform the analysis in the banking sector. On the other hand, Grohmann et al. (2018) relate financial literacy and inclusion, so they ignore the role of financial innovation and corruption. Similarly, Jungo et al. (2021) and Klapper and Lusardi (2020) do not consider the role of corruption when examining the effect of financial literacy and inclusion on bank profitability and financial stability. Given this, this study fills the existing gap in the literature on financial stability in the

banking sector by showing how financial literacy, financial innovation, and financial inclusion can reverse the adverse effect of corruption on financial stability.

The present study contributes to expanding the scarce literature that relates the role of institutional factors in corruption in the banking sector in several ways. First, the study simultaneously considers the role of financial literacy, financial innovation, and financial inclusion in the relationship between corruption and the financial stability of banks, which have been neglected by previous studies. Second, the study uses a fairly comprehensive sample consisting of 137 countries disaggregated by each continent to allow for comparative analysis across different continents, namely Africa, America, Europe, and Asia. The disaggregation of the sample was based on comparing the results due to the strong heterogeneity in the distribution of corruption by each continent, where African countries proved to be more corrupt and the population less financially literate. Third, based on the product of the indicator literacy and corruption and literacy and financial inclusion, we created two control variables, where the interaction between literacy and corruption serves to measure the ability of literacy to reverse the effect of corruption on our main variables of interest and the interaction between literacy and financial inclusion served to assess the complementarity between the two. So, the implementation of these variables and the consideration of literacy, innovation, and financial inclusion as mitigating factors of corruption constitute the originality of this study. Finally, we are unaware of any study that analyzes the factors that can mitigate the negative effect of corruption on the financial stability of banks.

Overall, our results confirm that increased corruption graces credit risk, reduces banks' profitability, and deteriorates banks' financial stability. Therefore, our results confirm that the negative effect of corruption can be reversed by financial literacy, innovation, and financial inclusion. Thus, as a practical implication, the study shows that financial innovation and inclusion translate into reducing the bureaucracy of access to financial products and services, which can reduce the opportunity for corrupt practices in accessing the banking sector while fostering financial literacy facilitates the rational use of scarce financial resources. Regarding the social implication, the study shows the need to create a mechanism that facilitates the access and use of financial services products, which can increase the welfare of households, banks, and the economy in general. The policy implication of the study is expressed in the need to foster innovation and inclusion and complement them with financial literacy in environments where corruption levels are high so that banks remain stable. The rest of the paper is organized in the next section by the literature review, followed by the data and methodology section. Results and discussions are presented in the fourth section, and in the fifth and last section, we present the conclusions, limitations, and implications of the study.

Literature Review

Corruption and Banking Stability

Banks are the main financial institutions in developing countries, and the savings of surplus agents are the main instruments of financial intermediation. Contrary to this,

in developed countries, financial systems are more diversified and complex, emphasizing the operation of other markets, such as the capital market. In this study, our attention will be focused on the effect of corruption on banking stability indicators, since banks play a key role in the effectiveness of financial inclusion programs. Thus, for banks to effectively play their role as financial intermediaries, they must be financially stable (Musau et al., 2018). Bank stability is the ability of a bank to efficiently and continuously intermediate financial transactions without any disruption, even in the event of internal or external shocks (Ali et al., 2019; Toader et al., 2018). Banking stability can be influenced by endogenous or exogenous factors; among the exogenous factors, the levels of corruption in the country and the degree of financial literacy are factors that deserve attention from everyone in the financial system.

Ali et al. (2019) found that the size of the bank, liquidity risk, and corruption have positive impacts on banking stability in Pakistan. In addition, the authors verified the existence of a negative relationship between credit risk and bank stability. Toader et al. (2018) showed that lower levels of corruption in emerging countries positively influence banking stability and reduce credit risk. The authors also found banks operating in countries that have adopted the code of corporate governance or European Union member countries are less affected by corruption. Yunan (2021) emphasized that corruption allows the allocation of financial resources to unproductive projects, so corruption negatively affects the profitability and soundness of Islamic banks. For the eurozone, Asteriou et al. (2021) found that corruption and lack of transparency have negative effects on bank profitability and stability. Some empirical evidence sustains that corruption negatively influences banking stability, through inefficiency in resource allocation and increased credit risk. Moreover, these studies show that the negative effect of corruption on banking stability is almost always indirect, through increased credit risk and reduced bank profitability, bribery, political connection, nepotism, or influence trait (for more details see for example the studies conducted by Athari & Bahraini, 2021; Ben Ali et al., 2020; Chen et al., 2018; Hewa Wellalage et al., 2020; Hung et al., 2017; Jenkins et al., 2021; Kamran et al., 2019; Liu et al., 2020; Mohamad & Jenkins, 2020; Rehman et al., 2020; Yakubu, 2019). In opposition, studies conducted by Bougatef (2015), Hanoteau et al. (2021), and Mohammad et al. (2019) report positive effects of corruption on bank profitability and stability. In short, the set of empirical literature presented considers only the negative effect of corruption on banking stability, but as far as we are aware, none of these studies identify the factors that can reverse these negative effects.

Literacy, Financial Inclusion, and Innovation, and Corruption and Banking Stability

Economic agents' knowledge levels and innovation are relevant factors for sustainable decision-making and contribute considerably to the economic progress of any economy (Nejjari & Aamoum, 2022). Education is one of the main forms of human capital accumulation, contributing significantly to increased productivity (Li & Chu, 2022). In the financial industry, high levels of financial education contribute to informed decision-making (Lusardi & Mitchell, 2011; Xu, 2012). Financial literacy is the ability

and familiarity with financial products and services, specifically the returns, costs, and associated risks (Jungo et al., 2021; Klapper & Lusardi, 2020). Bribes paid to obtain access to finance act as an extra cost of borrowing (Hanoteau et al., 2021); however, the acceptability of paying such a bribe may be conditioned by the borrower's financial knowledge or the levels of financial inclusion in the country. Financial inclusion provides an equal opportunity for agents to obtain financing, reducing inequality, but corruption sustains inequality among agents (Ouechtati, 2022). From this perspective, Ghardallou (2022) argues that access to finance and financial markets can be conditioned by political groups and other influential actors. Financial innovation enables greater access to financial products and services and is an excellent measure for fostering financial inclusion (Beck et al., 2015; Demirguc-Kunt et al., 2018).

Jungo et al. (2021) found that financial literacy and financial inclusion reduce credit risk, increase bank profits, and help underpin financial stability. Similarly, Klapper and Lusardi (2020) confirm that financial literacy increases bank resilience and improves stability.

Setor et al. (2021) showed that digital payments contribute to the improvement in the control of corruption, through the increase of transparency caused by the registration of the transaction and the reduction of contacts between agents. Analogously, Zhao et al. (2021) state that financial innovation reduces asymmetric information and transaction costs and allows banks to expand their business. Still, on this approach, Wang et al. (2021) showed that financial innovation improves banks' business model, reduces operating costs, strengthens risk control capability, and increases bank profitability, confirming the result presented in the study by Rega (2017). Financial inclusion is associated with easy access to financial products and services without any discrimination at a relatively low cost (Huang & Zhang, 2020; Zins & Weill, 2016), whereas corruption limits access to financial products and services. Terzi (2015) evidenced that financial inclusion increased the stability of banks in Turkey by boosting banks' deposit bases. Furthermore, Girón et al. (2021) add that financial inclusion increases savings. However, the studies conducted by Ahamed and Mallick (2019), Kouki (2020), and Musau et al. (2018) show that financial inclusion produces positive effects on banks' financial stability. Moreover, it is crucial to consider that financial inclusion and innovation contribute to the transformation of the economy from informal to formal and, in this way, reduces the amount of currency outside the financial system, which is also an important lubricant for corruption and money laundering, especially in developing countries (Ajide, 2020; Setor et al., 2021). Controlling corruption reduces inequality, and the existence of a strong informal sector is a strong barrier to controlling corruption (Khan, 2021). Sdiri and Ayadi (2021) evidenced that financial innovation contributes significantly to corruption control, furthermore, increased competitiveness in the banking sector contributes to corruption reduction. The scarce literature that highlights the relationship between corruption and banking stability does not directly consider the role of literacy, inclusion, and innovation as mitigating factors of the adverse effect of corruption on banking stability.

Methodology and Data

Data Description

Our study uses secondary data extracted from various data sources, and we set the period of our study to 2014 due to the availability of data for financial literacy. Thus, our cross-sectional sample consists of 137 countries, distributed in two (2) Oceania countries (included only in the full sample), thirty-five (35) African countries, thirty-nine (39) European countries, thirty-seven (37) Asian countries, and twenty-nine (29) American countries (see Table 10 in Appendix). The variables under study and their abbreviations are shown in Table 1. The data on financial literacy was extracted from the Standard & Poor's Ratings Services Global Financial Literacy Survey database, which examined financial literacy in over 140 countries in the year 2014, assessing the knowledge in over 150,000 adults, on the fundamental concepts regarding risk diversification, inflation, numeracy, and compound interest; therefore, the survey is a joint initiative of Mc Graw Hill Financial, Gallup, World Development Research Group and Global Financial Literacy. Authors such as Grohmann et al. (2018), Jungo et al. (2021), and Klapper and Lusardi (2020) have used the same data source in their studies. Adults are asked four questions on interest rate composition, the intertemporal value of money, risk diversification, and numeracy to measure their financial literacy. If they correctly answer three of the four questions, they are considered financially literate (Grohmann et al., 2018; Klapper & Lusardi, 2020).

Regarding data on financial innovation and financial inclusion, we used the World Bank-Global Financial Inclusion (Global Findex) database, which randomly surveys over 150,000 adults worldwide in three different years (2011, 2014, 2017), and to meet our objectives, we extracted only data from the year 2014. Studies with similar approaches were conducted by Feghali et al. (2021) and Setor et al. (2021), who used the same database.

The data on corruption was taken from two different sources. For the corruption perception index, we took data from the Transparency International database, being measured on a scale of zero (0–highly corrupt country) to one hundred (100–clean country), and the data on corruption control we obtained from the World Bank database (World Governance Indicators), being this indicator measured on the scale of−2.5 to 2.5, where higher values indicate less corruption and lower values are associated with greater corruption. Previous studies have used the same database to obtain corruption indicators (see Ajide, 2020; Jenkins et al., 2021; Mohamad & Jenkins, 2020).

The specific variables of banks such as credit risk, profitability, financial stability of banks, competitiveness, financial regulation, and operational efficiency, as well as the macroeconomic variables such as economic growth, were taken from the World Bank-Global Financial Development and Development Indicators database. It should be emphasized that also previous studies use the same database (Ahamed & Mallick, 2019; Albaity et al., 2019; Jungo et al., 2021; Mohammad et al., 2019; Wang et al., 2021).

Table 1 Studied variables: description and acronyms

Acronym	Description	Methodologically used as…
npl	Non-performing loans (credit risk)	Dep. Var. in FGLS
roa	Bank profitability	Dep. Var. in FGLS
zscore	Bank's financial stability (*z*-score of banks)	Dep. Var. in FGLS
cpi	Corruption perception index	Indep. Var. in FGLS
coc	Corruption control	Indep. Var. in FGLS
fl	Financial literacy	Indep. Var. in FGLS
atms	Percentage of adults over the age of 15 who used ATM to make payments	Constructed index Financial innovation index (using PCA) Independent variable in FGLS
Debcard	Percentage of adults over 15 who hold a debit card	
Credcard	Percentage of adults over 15 who hold a credit card	
netpabuy	Percentage of adults over age 15 who used the Internet to buy or pay for any good	
digpay	Percentage of adults over age 15 who have received or made digital payments in the past year	
bancount	Percentage of adults over 15 who have a bank account	Constructed index Financial inclusion index (using PCA) Independent variable in FGLS
Save	Percentage of adults over the age of 15 who made savings last year	
Borrowers	Percentage of adults over 15 years of age who have benefited from bank loans	
Deposit	Percentage of adults over 15 years who made bank deposits	
Withdraw	Percentage of adults over the age of 15 who conducted bank withdrawals at a financial institution	
banks	Bank branch for each 100,000 adults	
Credit	Bank credit as a percentage of the GDP	
flcpi	Product between financial literacy and corruption perception index	Control Var. in FGLS
flcoc	Product between financial literacy and control of corruption	Control Var. in FGLS
ififl	Product between inclusion and financial literacy	Control Var. in FGLS
Size	Bank size	Control Var. in FGLS
regf	Financial regulation	Control Var. in FGLS
GDP	Gross domestic product per capita growth rate	Control Var. in FGLS
iboone	Boone index (competitiveness)	Control Var. in FGLS

Existing studies on financial innovation have used different indicators of financial innovation, due to a lack of consensus on the existence of a single proxy, as well as, the lack of a specific database where these data can be extracted. In addition, information on the research and development of financial institutions is rarely collected and made available (Beck et al., 2015; Laeven et al., 2015). Therefore, it is known that as financial institutions innovate, agents give up more liquid assets to use less liquid assets (Chinoda & Kwenda, 2019; Dunne & Kasekende, 2018). Due to this, we used principal component analysis (PCA) as suggested by Qamruzzaman and Jianguo (2017) and Qamruzzaman and Wei (2019), to construct our financial innovation index, consisting of the following variables: percentage of adults aged 15 + who have used ATMs to make payments, who have a debit card and credit card, and who have used the Internet to buy or pay for some good and the percentage of adults aged 15 + who have received or made digital payments. To measure financial inclusion, we created the financial inclusion index through PCA as well, as suggested by previous studies conducted by Anarfo et al. (2019), Jungo et al. (2021), and Qamruzzaman and Wei (2019). We aggregated aspects related to geographic and demographic penetration of financial products and services in the inclusion index, respecting the multidimensionality of financial inclusion. Therefore, the variables that make up the index are the availability of bank branches per 100,000 adults, bank credit made available as a percentage of gross domestic product, and the percentage of adults over 15 years old who have a bank account, who have saved, withdrawn, or deposited/borrowed.

The empirical literature suggests using the ratio of non-performing loans over total loans to measure credit risk. While to measure financial stability the bank's *z*-score is used as a measure, since, it indicates the probability of bankruptcy of a bank and expresses the number of standard deviations a bank can fall before reaching bankruptcy (for more details on these measures see the study conducted by Feghali et al. (2021), Goetz (2018), and Klapper and Lusardi (2020)). To measure bank profitability, we use the returns on bank assets (Jenkins et al., 2021; Mohamad & Jenkins, 2020; Rehman et al., 2020). We use financial regulation, operational efficiency, competitiveness in the banking sector, bank size, and macroeconomic variable gross domestic product growth rate as control variables. Financial regulation is represented by regulatory capital by risk-weighted assets; however, financial regulation is expected to limit banks' risk avidity and contribute positively to improving banks' financial stability (Athari & Bahreini, 2021; Bougatef, 2015; Kamran et al., 2019). The operating efficiency of banks was measured by the ratio of operating costs to total revenue. Because of this, banks with efficient management exhibit low risk-taking behavior and are more profitable and profitability is positively associated with stability (Ben Khediri et al., 2009; Bougatef, 2015).

Regarding competitiveness, we use the Boone index, which measures competitiveness by the efficiency channel, relating the elasticity of profits and marginal costs. However, this indicator expresses that competitiveness improves the performance of more efficient banks and weakens the performance of inefficient banks (Albaity et al., 2019; Boone, 2008). A negative value of Boone's index indicates competition and the stronger the competition the higher the absolute value (Boone, 2008). To measure bank size, we use the bank capital to total asset ratio as a proxy. Bank size is a determining factor for obtaining market power and managing the

opportunities for diversification of assets and risks (Albaity et al., 2019). The macroeconomic environment is represented by the growth rate of gross domestic product; therefore, the economic conditions of the countries are expected to affect the supply and demand for financing and the ability of agents to repay loans (Gozgor, 2018; Mengistu & Saiz, 2018). Based on the product of the indicator literacy and corruption and literacy and financial inclusion we created two control variables, where the interaction between literacy and corruption serves to measure the ability of literacy to reverse the effect of corruption on our main variables of interest and the interaction between financial literacy and inclusion served to assess the complementarity between the two.

Model Specification

The data used in our study are composed of rates, indices, and integers. Thus, to eliminate any bias, we use natural logarithms in all variables so that we can standardize them. The specification of the index of inclusion and financial innovation was according to the suggestion of Anarfo et al. (2019) and is presented in Eqs. (1) and (2).

$$\begin{aligned} ifi_j = {} & W_{J1}banks_1 + W_{J2}bancount_2 + W_{J3}Save_3 + W_{J4}Borrowers_4 \\ & + W_{J5}Deposit_5 + W_{J6}Withdraw_6 + W_{J7}Credit_7 \end{aligned} \tag{1}$$

$$\begin{aligned} innov_p = {} & W_{p1}atms_1 + W_{p2}Debcard_2 + W_{p3}Credcard_3 \\ & + W_{p4}Debcard_4 + W_{p5}netpabuy_5 + W_{p6}digpay_6 \end{aligned} \tag{2}$$

where *ifi* and *innov* are the indices of inclusion and financial innovation, respectively, and W_J;W_P are the weights of the respective coefficients. The meaning of the variables can be found in Table 1. To infer the proper use of financial inclusion and financial innovation indexes, we performed the Kaiser–Meyer–Olkin test (KMO), as suggested by Carillo et al. (2019) and Carvalho (2013). The KMO test compares the simple correlations with partial correlations, in which the values of this statistic range from 0 to 1 and values close to 0 (zero) indicate that the use of the index may not be adequate and values close to 1 (one), indicating better adequacy in the use of the index (Carillo et al., 2019; Carvalho, 2013). The KMO test is specified according to Eq. (3).

$$KMO = \frac{\sum \sum_{j \neq k} r_{jk}^2}{\sum \sum_{j \neq k} r_{jk}^2 + \sum \sum_{j \neq k} q_{jk}^2} \tag{3}$$

where r_{jk}^2 represents the square of the correlation matrix of the original variables outside the diagonal and q_{jk}^2 is the square of the partial correlations between the variables.

We use the feasible generalized least squares (FGLS) model, which generates robust results in case of violation of classical assumptions, whose results are free of autocorrelation and heteroscedasticity problems; and it is a widely used model in cross-sectional data (Greene, 2018; Miller, 2017; Saha et al., 1997; Umoru & Osemwegie, 2016).

$$Y_{it} = X_{it}\beta_{it} + u \tag{4}$$

where Y_{it} is the vector of k dependent variables such as credit risk (npl), profitability and financial stability of banks (z-score of banks). The X_{it} is the matrix of the explanatory variables of the model, namely, corruption, financial literacy, financial innovation, financial inclusion, and the other control variables. The β_{it} are the parameters of the vectors of the explanatory and control variables, and u is the vector of random errors.

Results and Discussions

The results and related discussions will be presented in each subsection. Regarding the term of interaction between financial literacy and corruption, financial literacy and financial inclusion are the novelty of this study, so we will have little possibility to compare with previous studies because they do not employ this approach.

Descriptive Statistics

Table 2 presents the summary of the average behavior of the variables included in the study, for all samples. Complete information on descriptive statistics can be found in Tables 11, 12, and 13 in Appendix. From the results of the descriptive statistics for the corruption indicators (control of corruption and corruption perception index), we can see that on average African countries are the most corrupt, followed by Asian and American countries, while European countries have the most controlled levels of corruption. No other continent country was included in the sample due to a lack of reasonable data to fulfill the study in comparison terms.

Regarding financial literacy, we can see that on average the population of European countries has higher levels of financial literacy compared to the population of African, Asian, and American countries. For the financial inclusion indicators, there is a big difference in the average number of bank branches available for the African population and the rest of the world. That is, we find that on average for every 100,000 adults in African countries, there are only 7 bank branches, in Asian countries around 16 bank branches, in American countries around 17 bank branches, and in European countries around 31 bank branches. These results indicate that there is a strong demographic penetration of bank branches in European countries and a weak one in African countries.

Similar average behavior can be seen in the other indicators of financial inclusion such as the percentage of adults over the age of 15 who said that they have a bank account and that they make bank deposits and savings, as well as, the percentage of bank credit granted by the financial system, which is lower in African countries and higher in European countries.

Regarding financial innovation, we find that on average the adult population over 15 years old that uses a debit card in Africa is only 15%, in Asia 34.8%, in America 36.4%, and in Europe about 64.6%. For credit cards, the average

Table 2 Summary of the mean behavior of the variables included in the study for all samples

	World	Africa	Asia	America	Europe
	Mean	Mean	Mean	Mean	Mean
coc	2.433	1.814	2.192	2.338	3.139
cpi	0.435	0.327	0.370	0.430	0.571
Fl	0.367	0.343	0.344	0.355	0.399
bank	16.818	7.108	15.567	17.410	30.433
bancount	0.550	0.309	0.525	0.512	0.780
Save	0.230	0.128	0.216	0.203	0.322
credit	3.586	30.149	57.959	49.845	78.800
Borrowers	0.124	0.059	0.140	0.138	0.154
Deposit	0.774	0.739	0.723	0.736	0.867
Withdraw	0.781	0.724	0.714	0.775	0.886
atms	51.124	15.152	48.577	52.229	79.742
Debcard	0.396	0.150	0.348	0.364	0.646
Credcard	0.180	0.033	0.175	0.196	0.307
netpabuy	0.175	0.025	0.124	0.120	0.360
digpay	0.438	0.232	0.376	0.384	0.677
GDP	2.283	2.505	2.915	1.532	2.040
npl	9.356	12.634	8.850	3.231	11.012
size	11.893	10.656	13.936	9.662	12.782
eficiency	55.380	56.124	51.368	56.631	57.528
regf	16.918	17.288	16.651	15.656	17.722
roa	1.495	1.819	1.850	1.618	0.870
zscore	13.907	13.077	15.478	14.842	12.057
iboone	−0.035	−0.025	0.019	−0.066	−0.068

See variables' acronyms and definitions in Table 1

behavior is identical, where only 3.3% of the population uses a credit card in Africa, while in Asian and American countries it is 17.5% and 19.6%, respectively, whereas for European countries about 30.7% of the population uses a credit card. In African countries, only 2.5% of the adult population over 15 years old used the Internet to buy or pay for some good, in Asian countries 12.4%, in American countries 12%, and in European countries about 36% of adults who used the Internet to pay or buy some goods in the period under analysis.

On average, credit risk is higher in African countries than in the rest of the world. Financial regulation is more pronounced in African and European countries than in Asian and American countries. Interestingly, we find that on average the profitability of banks is higher in African and Asian countries, while on average banks in Asian and European countries are more stable. Regarding competitiveness, we find that on average the banking sector in European and American countries is more competitive.

KMO Adequacy Test

The KMO goodness of fit test (Table 3) suggests the use of the inclusion and financial innovation indices in all samples because they have values greater than 50%. For more details, check Carillo et al. (2019) and Carvalho (2013).

Correlation Matrix

The results of the correlation matrix can be found in Table 14 in Appendix. The results suggest that there is no evidence of multicollinearity problems among the variables included in the study, except for the corruption indicators, specifically the corruption perception index and control of corruption, which show a positive, strong, and statistically significant correlation, above 70%.

Corruption and Credit Risk

Results presented in Table 4, by application of the FGLS model, indicate that corruption increases credit risk in Asian countries. Besides, the robustness of our results is guaranteed by the use of an alternative indicator of corruption (corruption control in Table 5) in all estimations. Therefore, for African, American, and European countries, we do not find statistical significance, so we can conclude nothing concerning the effect of corruption on credit risk in these countries. This result regarding the positive effect of corruption on credit risk is in line with the results presented in previous studies on samples from Asian banks and other regions (Gozgor, 2018; Jenkins et al., 2021; Mohamad & Jenkins, 2020; Rehman et al., 2020).

Moreover, increased financial literacy in Asian countries reduces credit risk. This result is confirmed in the estimation of the global sample, but for American countries, financial literacy was not able to reverse the effect of corruption. Previous studies presented by Jungo et al. (2021) and Klapper and Lusardi (2020) confirm that financial literacy reduces credit risk. Increased financial inclusion reduces credit risk in African and Asian countries, coinciding with the results presented by Musau et al. (2018) when they found that financial inclusion reduces credit risk in Kenya.

Table 3 Factor adequacy test for all samples

	Africa		**Asia**		**America**		**Europe**		**World**	
Variables	ifi	innov	ifi	innov	ifi	innov	ifi	innov	ifi	innov
CM	0.011	0.035	0.009	0.005	0.016	0.001	0.09	0.002	0.018	0.002
Chi^2	138	106	155.8	179	78.26	143.4	82.3	217	536.10	809.63
P-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
KMO	**0.57**	**0.75**	**0.643**	**0.84**	**0.69**	**0.78**	**0.74**	**0.80**	**0.72**	**0.86**

The KMO values for the financial inclusion and financial innovation indicators are shown in bold

CM correlation matrix, *KMO* Kaiser–Meyer–Olkin, *ifi* financial inclusion indicator, *innov* financial innovation indicator

Table 4 The effect of corruption on credit risk

	World	Africa	Asia	America	Europe
	npl	npl	npl	npl	npl
cpi	0.550	0.177	1.976**	−0.695	−1.787
	(0.203)	(0.922)	(0.020)	(0.578)	(0.305)
fl	−5.956*	−0.725	−7.432***	6.648*	5.407
	(0.057)	(0.900)	(0.003)	(0.076)	(0.380)
ifi	−0.276	−1.710**	−0.881*	−0.171	0.0390
	(0.310)	(0.045)	(0.072)	(0.744)	(0.932)
innov	−0.320***	0.604	−0.251	−0.994***	−0.198
	(0.007)	(0.183)	(0.185)	(0.003)	(0.496)
eficiency	−0.168	−0.248	−0.448	0.568	−0.0385
	(0.618)	(0.632)	(0.470)	(0.367)	(0.956)
regf	−0.146	0.540	0.0976	3.498**	0.113
	(0.375)	(0.291)	(0.672)	(0.018)	(0.619)
iboone	0.761*	2.354	0.288	−1.375	0.430
	(0.077)	(0.270)	(0.796)	(0.383)	(0.347)
size	−0.157	−0.473	0.659	−1.575**	0.594
	(0.344)	(0.222)	(0.148)	(0.026)	(0.168)
GDP	−0.0961	−0.0388	−0.179*	−0.302***	−0.214
	(0.201)	(0.821)	(0.090)	(0.004)	(0.241)
flcpi	0.591**	0.363	2.209***	−1.864*	−1.344
	(0.047)	(0.818)	(0.002)	(0.051)	(0.372)
ififl	−0.157	−0.950*	−0.665*	−0.144	0.243
	(0.448)	(0.080)	(0.069)	(0.735)	(0.524)
_cons	5.619***	2.867	−5.076	−5.073	7.656
	(0.003)	(0.653)	(0.261)	(0.475)	(0.294)

p-values in parentheses; * $p<0.10$, ** $p<0.05$, and *** $p<0.01$. See Table 1 for variables' acronyms

The presented results also seem to indicate that financial innovation reduces credit risk in American countries and the global sample, analogous results can be seen in Zhao et al. (2021) and Sdiri and Ayadi (2021), reasoning that financial innovation reduces asymmetric information and allows banks to better monitor their customers. Financial regulation increases credit risk in American countries, meaning that banks are more risk-averse in these countries. These results can be justified by the average behavior of these variables presented through descriptive statistics (Table 2), where financial liberalization levels vary by 17.72 for European countries, 17.28 for African countries, 16.65 for Asian countries, and only 15.66 for American countries (higher financial liberalization).

Additionally, higher rigidity of financial regulations preserves stability by preventing unsustainable credit growth (Anarfo et al., 2020; Asteriou et al., 2021). Also, bank size significantly reduces credit risk, for this group of countries. This result indicates that larger banks have greater resources and more efficient techniques for selecting good customers (Albaity et al., 2019).

Table 5 Verification of robustness for credit risk

	Mundo	Africa	Asia	America	Europa
	npl	npl	npl	npl	npl
coc	0.764	−0.279	1.998*	−2.030	−2.073
	(0.343)	(0.892)	(0.064)	(0.105)	(0.278)
fl	−2.206	2.038	−1.575**	2.197**	1.702
	(0.202)	(0.632)	(0.024)	(0.046)	(0.379)
ifi	−0.366	−1.728**	−0.887*	−0.00941	0.0503
	(0.216)	(0.036)	(0.069)	(0.986)	(0.911)
innov	−0.261**	0.607	−0.123	−0.625**	−0.176
	(0.044)	(0.171)	(0.566)	(0.025)	(0.565)
eficiency	−0.288	−0.218	−0.393	0.464	−0.0222
	(0.401)	(0.656)	(0.512)	(0.478)	(0.975)
regf	−0.102	0.565	0.0705	1.445	0.120
	(0.536)	(0.259)	(0.759)	(0.196)	(0.593)
iboone	0.725*	2.501	0.379	−0.227	0.393
	(0.091)	(0.238)	(0.731)	(0.890)	(0.385)
size	−0.0684	−0.502	0.462	−0.596	0.591
	(0.693)	(0.169)	(0.319)	(0.242)	(0.165)
GDP	−0.0659	0.0135	−0.153	−0.235**	−0.209
	(0.390)	(0.951)	(0.149)	(0.023)	(0.254)
flcoc	0.984	−0.0468	2.809***	−2.659**	−1.518
	(0.130)	(0.978)	(0.002)	(0.014)	(0.341)
ififl	−0.255	−0.948*	−0.689*	−0.0225	0.248
	(0.264)	(0.072)	(0.059)	(0.959)	(0.504)
_cons	4.302***	2.160	0.883	−1.755	2.711
	(0.005)	(0.419)	(0.781)	(0.682)	(0.437)

p-values in parentheses; * $p<0.10$, ** $p<0.05$, and *** $p<0.01$. See Table 1 for variables' acronyms. Now corruption is controlled by coc (corruption control)

The interaction between corruption and financial literacy causes positive effects on credit risk in Asian countries, meaning that an increase in financial literacy is insignificant when corruption in the banking sector is systematic. However, the interaction between financial literacy and corruption in American countries reduces credit risk considerably, meaning that in the case of the existence of corruption in lending, and the borrower has financial skill and competencies; the probability of default reduces. Moreover, the interaction between inclusion and financial literacy produces a negative effect on credit risk in Asian countries. These results emphasize the complementarity of these two variables for credit risk reduction, where the study by Jungo et al. (2021) confirms that financial literacy complements financial inclusion for credit risk reduction.

Corruption and Profitability of Banks

The results presented in Table 6, confirmed in Table 7, reveal that there is no statistical significance between corruption and profitability in African, Asian, and European countries. Therefore, these results suggest that corruption reduces the profitability of banks in American countries, being supported by the study of Asteriou et al. (2021) and Yunan (2021), which conclude that corruption leads to inefficient allocation of credit, increasing default, and reducing profitability. In addition, the results suggest that financial literacy, financial inclusion, and financial innovation increase bank profitability in this group of countries. These results suggest that better utilization of loans due to financial literacy can contribute to the increase of banks' profit. Moreover, financial inclusion increases banks' deposit portfolios, expanding the availability of resources to lend (Jungo et al., 2021; Klapper & Lusardi, 2020; Kumar et al., 2021).

Table 6 Corruption effect over profitability (ROA (return on assets))

	World	Africa	Asia	America	Europe
	roa	roa	roa	roa	roa
cpi	−0.196	2.760	−0.439	−2.403***	0.00360
	(0.635)	(0.227)	(0.293)	(0.000)	(0.999)
fl	1.714	−11.02	1.322	4.695***	−1.906
	(0.566)	(0.132)	(0.281)	(0.001)	(0.848)
ifi	−0.431*	0.789	−0.274	0.379*	0.450
	(0.097)	(0.465)	(0.254)	(0.054)	(0.538)
innov	−0.163	−0.409	−0.0538	0.379***	0.427
	(0.153)	(0.475)	(0.563)	(0.002)	(0.361)
eficiency	−0.950***	−1.432**	−0.803***	−0.337	−1.383
	(0.003)	(0.029)	(0.008)	(0.154)	(0.219)
regf	0.353**	0.0821	0.653***	−4.049***	0.415
	(0.024)	(0.899)	(0.000)	(0.000)	(0.258)
iboone	0.611	−4.892*	1.679***	1.910***	0.396
	(0.137)	(0.070)	(0.002)	(0.001)	(0.591)
size	0.262*	0.374	0.478**	1.921***	1.211*
	(0.098)	(0.445)	(0.032)	(0.000)	(0.081)
GDP	−0.0374	−0.123	0.0134	−0.0912**	0.398
	(0.603)	(0.571)	(0.796)	(0.021)	(0.175)
flcpi	−0.125	2.924	−0.349	−1.059***	0.508
	(0.660)	(0.143)	(0.326)	(0.003)	(0.834)
ififl	−0.161	0.282	−0.0128	0.353**	0.324
	(0.414)	(0.681)	(0.943)	(0.027)	(0.598)
_cons	1.914	−3.622	2.131	18.40***	0.953
	(0.290)	(0.653)	(0.336)	(0.000)	(0.935)

p-values in parentheses; * $p<0.10$, ** $p<0.05$, and *** $p<0.01$. See Table 1 for variables' acronyms

Table 7 Robustness checks for profitability

	World	Africa	Asia	America	Europe
	roa	roa	roa	roa	roa
coc	0.470	2.254*	−0.856	−1.233*	1.321
	(0.540)	(0.067)	(0.101)	(0.066)	(0.668)
fl	−0.698	−1.624**	0.472	0.845	−0.858
	(0.671)	(0.048)	(0.160)	(0.152)	(0.783)
ifi	−0.513*	0.452	−0.243	0.201	0.212
	(0.068)	(0.385)	(0.303)	(0.493)	(0.771)
innov	−0.173	−0.0864	0.0176	−0.0149	0.178
	(0.160)	(0.745)	(0.865)	(0.920)	(0.718)
eficiency	−1.001***	−1.453***	−0.809***	−0.305	−1.354
	(0.002)	(0.000)	(0.005)	(0.385)	(0.230)
regf	0.350**	0.611*	0.689***	−1.963***	0.469
	(0.026)	(0.051)	(0.000)	(0.001)	(0.194)
iboone	0.630	−7.279***	1.789***	1.009	0.352
	(0.122)	(0.000)	(0.001)	(0.253)	(0.630)
size	0.295*	−0.230	0.397*	0.883***	1.291*
	(0.074)	(0.353)	(0.077)	(0.001)	(0.060)
GDP	−0.0425	−0.00272	0.0269	−0.167***	0.288
	(0.560)	(0.980)	(0.599)	(0.003)	(0.329)
flcoc	0.460	2.219**	−0.488	−0.530	0.892
	(0.457)	(0.037)	(0.277)	(0.362)	(0.729)
ififl	−0.233	0.410	0.00296	0.205	0.185
	(0.284)	(0.225)	(0.987)	(0.385)	(0.757)
_cons	2.710*	3.013**	1.347	6.552***	−0.869
	(0.302)	(0.012)	(0.381)	(0.004)	(0.877)

p-values in parentheses; * $p<0.10$, ** $p<0.05$, and *** $p<0.01$. See Table 1 for variables' acronyms. Now corruption is controlled by coc (corruption control)

Operational inefficiency significantly reduces bank profitability in African and Asian countries, as evidenced by Ben Khediri et al. (2009). Financial regulation increases profitability in Asian banks, whereas for banks in American countries financial regulation reduces profitability. We believe that this occurs because the banking systems in American countries are more flexible, and easily banks can take on too much risk. Authors such as Ben Ali et al. (2020) and Yakubu (2019) found that regulation produces positive effects on profitability.

The results also show that increasing competitiveness reduces the profitability of banks in African countries, however, in Asian and American countries increasing competitiveness improves the profitability of banks. Parallel to these results, Goetz (2018) confirms that increasing competitiveness produces positive effects on profitability in the United States. Regarding the effect of bank size on profitability, our results suggest that there is no statistical significance in African countries. However, we find that bank size is a relevant factor in increasing bank profitability in Asian, American, and European countries, with larger banks having better resources to

assess the creditworthiness of each customer, better diversify risk, and take advantage of scale gains (Albaity et al., 2019; Yakubu, 2019).

The interaction term between corruption and financial literacy produces negative effects on the profitability of banks in American countries. Thus, it is understood that the financial literacy of the population in American countries is not strong enough to alleviate the negative effect of corruption on profitability. Additionally, we find that the interaction term between inclusion and financial literacy increases bank profitability in Asian countries, meaning that expanding access to financial services and products to all members of the population and increasing the financial literacy levels of the population included in the financial system can contribute to increasing bank profitability, thus leading to higher safety, stability, and trust in the banking system.

Corruption and Financial Stability

The results presented in Table 4 suggest that we can conclude that there is nothing about the effect of corruption on the samples of African, American, and European countries because there is no statistical significance. However, the same results also suggest that corruption produces positive effects on the financial stability (measured through the *z*-score) of Asian banks (Table 8). These results support the evidence found by Ben Ali et al. (2020) and Trabelsi & Trabelsi, 2020), which showed that corruption can benefit banks by expanding access to financing through bribery. But even if this occurs, it will only be beneficial in the short term, and over time, it may increase defaults since bribery acts as an extra cost associated with loans. The results show that increased financial inclusion positively impacts financial stability in banks in Asian countries by 54.2% and in European countries by 80.9%. These results indicate that increased financial inclusion in Asian and European countries contributes to a reduction in the probability of bank failure. Besides, the difference in the size of the impact can be justified by the results of the descriptive statistics (see Table 2), concerning the indicators of financial inclusion that are much higher in European countries. Similar results can be found in Ahamed and Mallick (2019) and Kouki (2020).

The results also indicate that innovation produces positive effects on the stability of Asian banks. These results are also found by Cheng and Qu (2020) and Rega (2017), justifying that financial innovation improves the bank's business model, increases customer monitoring, and increases bank profits. However, our results contradict the results found by Li et al. (2021) and Zhao et al. (2021), which demonstrate that financial innovation reduces banks' profits through increased costs.

Additionally, we find that the interaction term between financial literacy and corruption produces a positive effect on financial stability in Asian banks. This result suggests that literacy reverses the negative effect of corruption on bank stability when the borrower has high levels of financial literacy. This result supports the arguments presented by Lalountas et al. (2011) and later reinforced by Asteriou et al. (2021), regarding the ability of corruption to enable increased access to credit in countries with high-risk aversion, causing a positive effect on banks' loan portfolio expansion and profitability. We also find that the interaction between financial inclusion and financial literacy produces positive effects on financial stability in Asian

Table 8 The effect of corruption on financial stability (*z*-score)

	World	Africa	Asia	America	Europe
	*z*score	zscore	zscore	zscore	zscore
cpi	0.0714	−1.098	1.553***	−1.500	−0.915
	(0.771)	(0.327)	(0.002)	(0.237)	(0.282)
fl	1.477	2.533	−3.240**	2.502	3.770
	(0.406)	(0.481)	(0.029)	(0.511)	(0.210)
ifi	0.229	0.348	0.542*	−0.0260	0.809***
	(0.138)	(0.512)	(0.062)	(0.961)	(0.000)
innov	−0.0187	−0.186	0.199*	0.446	0.220
	(0.783)	(0.510)	(0.077)	(0.181)	(0.120)
lgeficiency	−0.0393	−0.644**	0.169	−0.236	0.248
	(0.837)	(0.046)	(0.645)	(0.712)	(0.467)
regf	−0.122	−0.240	−0.324**	−1.154	−0.284**
	(0.190)	(0.450)	(0.018)	(0.440)	(0.011)
iboone	−0.382	−2.141	−1.493**	−0.805	0.0315
	(0.118)	(0.106)	(0.024)	(0.614)	(0.888)
size	0.0657	0.279	−0.169	0.196	−0.362*
	(0.486)	(0.247)	(0.530)	(0.785)	(0.085)
GDP	0.0138	−0.162	−0.0833	0.247**	0.147*
	(0.746)	(0.129)	(0.182)	(0.021)	(0.100)
flcpi	−0.142	−0.801	0.974**	−0.805	−0.904
	(0.400)	(0.414)	(0.023)	(0.406)	(0.218)
ififl	0.259**	0.150	0.446**	0.176	0.618***
	(0.028)	(0.656)	(0.040)	(0.684)	(0.001)
_cons	1.414	8.621**	−2.079	11.21	6.628*
	(0.189)	(0.029)	(0.437)	(0.120)	(0.063)

p-values in parentheses; * $p<0.10$, ** $p<0.05$, and *** $p<0.01$. See Table 1 for variables' acronyms

and European countries. Therefore, these results justify the need for complementarity between financial inclusion and financial literacy to preserve financial stability, even in countries with high levels of corruption. These results are evidenced as well in Table 9, providing a robustness check with a different measure of corruption control.

Financial regulation, specifically, the capital adequacy ratio, produces negative effects on financial stability in banks in Asian and European countries. This means that greater flexibility of financial regulations in these countries may reduce banks' ability to provide financial services and products and consequently become more unstable due to greater risk-taking. This result is similar to that found by Li et al. (2021), who conclude that weak financial regulation levels negatively impact banks' profits, supporting the result found earlier by Bougatef (2015) who gauged that financial regulation is important for banks' soundness.

The results also suggest that increased competitiveness in the Asian banking sector reduces bank stability, while in European countries financial stability is negatively impacted by bank size. In African countries, banks are negatively impacted by

Table 9 Verification of robustness for financial stability (*z*-score)

	World	Africa	Asia	America	Europe
	zscore	zscore	zscore	zscore	zscore
coc	0.0839	1.502	2.127***	−1.524	−1.149
	(0.852)	(0.224)	(0.001)	(0.192)	(0.217)
fl	0.749	−4.796*	−0.668*	0.151	1.277
	(0.439)	(0.060)	(0.095)	(0.883)	(0.175)
ifi	0.236	0.0560	0.456	−0.0382	0.828***
	(0.153)	(0.910)	(0.105)	(0.940)	(0.000)
innov	−0.0761	−0.00278	0.0625	0.436*	0.246*
	(0.294)	(0.992)	(0.613)	(0.093)	(0.099)
eficiency	−0.00338	−0.724**	0.0537	−0.234	0.251
	(0.986)	(0.014)	(0.876)	(0.701)	(0.461)
regf	−0.153*	−0.490	−0.400***	−1.018	−0.291***
	(0.098)	(0.103)	(0.003)	(0.329)	(0.008)
iboone	−0.379	−2.051	−1.598**	−0.970	0.0286
	(0.113)	(0.107)	(0.012)	(0.527)	(0.897)
size	0.0501	0.587***	−0.0484	0.129	−0.380*
	(0.605)	(0.007)	(0.857)	(0.786)	(0.067)
GDP	−0.00826	−0.307**	−0.116*	0.244**	0.158*
	(0.847)	(0.019)	(0.058)	(0.012)	(0.075)
flcoc	−0.333	1.292	1.101**	−0.779	−1.017
	(0.359)	(0.205)	(0.040)	(0.441)	(0.191)
ififl	0.279**	−0.0291	0.397*	0.183	0.624***
	(0.029)	(0.927)	(0.060)	(0.656)	(0.001)
_cons	2.142**	7.131***	2.163	6.645*	4.278**
	(0.012)	(0.000)	(0.238)	(0.096)	(0.012)

p-values in parentheses; * $p<0.10$, ** $p<0.05$, and *** $p<0.01$. See Table 1 for variables' acronyms. Now corruption is controlled by coc (corruption control)

operational efficiency, indicating that cost structure is a relevant variant to achieve stability of African banks, as evidenced by Ben Khediri et al. (2009). Thus, management efficiency increases bank profit and financial stability. Regarding the macroeconomic conditions of the countries, we find that increases in the economic growth rate have positive effects on the financial stability of banks in American and European countries. These results end up supporting the result of Gozgor (2018) who showed that better macroeconomic conditions such as higher productivity, less poverty, and more jobs positively affect the supply of credit and the ability to repay customers.

Conclusion

This study is the first to consider the ability of literacy, innovation, and financial inclusion to mitigate the negative effect of corruption on bank performance indicators across different countries and continents. Banking sector stability is a necessary condition for economic

stability and is one of the sources of sustainable and inclusive growth (Ben Ali et al., 2020; Trabelsi & Trabelsi, 2020). The performance of banks does not only depend on the internal factors of the banking sector but also on regulatory quality, country legal, political, and economic stability, customer demand, and financial skills which can influence the performance of banks (Ben Ali et al., 2020; Feghali et al., 2021).

Overall, our study confirms that corruption increases credit risk and reduces bank profitability and bank stability. Therefore, these results support the evidence found previously in other studies (Asteriou et al., 2021; Athari & Bahreini, 2021; Chen et al., 2018; Hewa Wellalage et al., 2020; Hung et al., 2017; Jenkins et al., 2021; Kamran et al., 2019; Mohamad & Jenkins, 2020; Rehman et al., 2020; Yakubu, 2019). Additionally, we find that financial literacy, financial inclusion, and financial innovation produce positive effects on bank profitability and stability and allow for the reduction of credit risk. Moreover, our results indicate that financial literacy and financial inclusion can be used as mitigating factors for the effect of corruption on banking stability indicators. In terms of comparing the results for each sample group, we found the effect and the size of this effect of our main variables were not the same, so these differences may be justifiable by the heterogeneity in the distribution of corruption in each group of countries. Thus, it would be interesting to evaluate the heterogeneous effect of financial literacy, financial inclusion, and financial innovation on the relationship between corruption and financial stability in the future.

Meanwhile, our study contributes to the growing empirical literature on economics and finance by considering three relevant topics (financial literacy, financial inclusion, and financial innovation) that are, in fact, a problem in emerging and developing countries, which can be used to improve financial access (financial inclusion and financial innovation) and monetize financial resources (financial literacy) while contributing to controlling corruption and fostering financial stability.

This study produces important policy and practical implications for policymakers, indicating that to achieve two major economic objectives, such as combating corruption and fostering financial stability, it is critical to consider financial inclusion, primarily through innovation and increasing the financial literacy of those included. Financial literacy can contribute to better profitability of scarce resources, better diversification, and rational decision-making. On the other hand, all measures aimed at fostering inclusion and literacy may have multiple positive effects, reducing poverty, and inequality, improving financial stability, and better-controlling corruption.

The limitation of our study is that we fixed the data to the 2014 period, due to the availability of financial literacy data for several countries. We are unaware of any other database that assesses financial literacy levels across more than 100 countries and with a higher period (not allowing a panel data study). So we were not able to highlight the attenuating effect of financial literacy, inclusion, and innovation over time. If a country-level database exists with more than one period, it would be imperative to use it in future studies. A comparative analysis among countries and time would also be possible since different policies pursued would help to explain some of the heterogeneities here highlighted by a continent/region.

Appendix

Table 10 Countries that compose the sample

Afghanistan	Cambodia	Germany	Kyrgyz Republic	Norway	Sweden
Albania	Cameroon	Ghana	Latvia	Pakistan	Switzerland
Algeria	Canada	Guatemala	Lebanon	Panama	Taiwan, China
Angola	Chad	Guinea	Lithuania	Peru	Tajikistan
Argentina	Chile	Haiti	Luxembourg	Philippines	Tanzania
Armenia	China	Honduras	Malawi	Poland	Thailand
Australia	Colombia	Hong Kong SAR, China	Malaysia	Portugal	Togo
Austria	Congo, Dem	Hungary	Mali	Puerto Rico	Tunisia
Azerbaijan	Congo, Rep	India	Malta	Romania	Turkey
Bahrain	Costa Rica	Indonesia	Mauritania	Russian	Turkmenistan
Bangladesh	Côte d'Ivoire	Iran, Islamic Rep	Mauritius	Rwanda	Uganda
Belarus	Croatia	Iraq	Mexico	Saudi Arabia	Ukraine
Belgium	Cyprus	Ireland	Moldova	Senegal	United Arab Emirates
Belize	Czech Republic	Israel	Mongolia	Serbia	United Kingdom
Benin	Denmark	Italy	Montenegro	Sierra Leone	United States
Bhutan	Dominican Republic	Jamaica	Myanmar	Singapore	Uruguay
Bolivia	Ecuador	Japan	Namibia	Slovak Republic	Uzbekistan
Bosnia and Herzegovina	Egypt, Arab Rep	Jordan	Nepal	Slovenia	Venezuela, RB
Botswana	El Salvador	Kazakhstan	Netherlands	Somalia	Vietnam
Brazil	Finland	Kenya	New Zealand	South Africa	Yemen, Rep
Bulgaria	France	Korea, Rep	Nicaragua	Spain	Zambia
Burkina Faso	Gabon	Kosovo	Niger	Sri Lanka	Zimbabwe
Burundi	Georgia	Kuwait	Nigeria	Sudan	

Table 11 Descriptive statistics for the sample of African countries and the world

	World				Africa					
	obs	mean	sd	min	obs	max	mean	sd	min	max
coc	137	2.433	1.052	0.84	35	4.756	1.814	0.628	0.84	3.355
cpi	136	0.435	0.205	0.08	35	0.920	0.327	0.118	0.08	0.630
fl	137	0.367	0.139	0.13	35	0.710	0.343	0.114	0.13	0.660
bank	137	16.818	16.173	0.04	35	79.570	7.108	5.748	0.79	22.470
bancount	137	0.550	0.301	0.06	35	1.000	0.309	0.197	0.07	0.822
Save	137	0.230	0.189	0.01	35	0.784	0.128	0.095	0.02	0.355
credit	137	3.586	1.127	0.03	35	5.522	30.149	33.968	0.03	150.974
Borrowers	137	0.124	0.076	0.04	35	0.405	0.059	0.042	0.01	0.171
Deposit	137	0.774	0.132	0.28	35	0.987	0.739	0.078	0.58	0.899
Withdraw	137	0.781	0.133	0.30	35	0.990	0.724	0.077	0.55	0.880
atms	137	51.124	47.051	0.47	35	280.808	15.152	18.130	0.47	65.500
Debcard	137	0.396	0.304	0.01	35	0.986	0.150	0.157	0.01	0.620
Credcard	137	0.180	0.199	0.01	35	0.771	0.033	0.039	0.03	0.173
netpabuy	137	0.175	0.217	0.02	35	0.790	0.025	0.017	0.02	0.076
digpay	137	0.438	0.300	0.028	35	0.992	0.232	0.163	0.04	0.691
GDP	137	2.283	2.252	−4.73	35	8.283	2.505	1.567	−0.38	6.103
npl	137	9.356	11.892	0.39	35	68.777	12.634	15.156	0.76	68.777
size	137	11.893	9.300	0.28	35	73.714	10.656	3.687	0.28	16.524
eficiency	137	55.380	13.011	9.11	35	90.967	56.124	14.254	9.11	82.954
regul	137	16.918	7.515	−1.94	35	62.242	17.288	5.185	3.55	30.210
roe	137	11.492	9.176	−18.48	35	56.773	14.507	7.410	−2.38	37.636
zscore	137	13.907	8.152	1.374	35	51.132	13.077	6.672	3.47	32.698
iboone	137	−0.035	0.216	−1.319	35	0.978	−0.025	0.092	−0.13	0.294

See Table 1 for the acronyms' description

Table 12 Descriptive statistics for the sample of Asian and American countries

	Asia					America				
	obs	mean	sd	min	max	obs	mean	sd	min	max
coc	37	2.192	0.953	0.945	4.572	23	2.338	0.916	1.100	4.337
cpi	37	0.370	0.189	0.080	0.840	22	0.430	0.183	0.190	0.810
fl	37	0.344	0.136	0.140	0.710	23	0.355	0.136	0.180	0.680
bank	37	15.567	13.288	1.680	71.230	23	17.410	8.663	2.700	35.950
bancount	37	0.525	0.307	0.064	0.966	23	0.512	0.209	0.189	0.991
Save	37	0.216	0.171	0.009	0.604	23	0.203	0.136	0.041	0.626
credit	37	57.959	51.710	0.436	233.211	23	49.845	31.378	13.824	136.851
Borrowers	37	0.140	0.091	0.004	0.405	23	0.138	0.057	0.020	0.270
Deposit	37	0.723	0.154	0.282	0.959	23	0.736	0.124	0.433	0.974
Withdraw	37	0.714	0.147	0.302	0.965	23	0.775	0.129	0.411	0.968
atms	37	48.577	50.786	0.750	280.808	23	52.229	43.340	1.090	221.854
Debcard	37	0.348	0.280	0.017	0.894	23	0.364	0.220	0.041	0.932
Credcard	37	0.175	0.248	0.001	1.000	23	0.196	0.184	0.035	0.771
netpabuy	37	0.124	0.139	0.002	0.525	23	0.120	0.174	0.009	0.657
digpay	37	0.376	0.287	0.028	0.887	23	0.384	0.215	0.104	0.963
GDP	37	2.915	2.959	−3.980	8.283	23	1.532	2.254	−4.732	5.819
npl	37	8.850	11.003	0.391	57.436	23	3.231	3.600	0.518	18.064
size	37	13.936	11.261	4.784	51.107	23	9.662	2.627	4.943	14.880
eficiency	37	51.368	12.995	24.584	80.384	23	56.631	10.570	38.543	75.707
regul	37	16.651	9.669	0.818	62.242	23	15.656	1.608	12.300	19.069
roa	37	1.850	3.419	0.313	21.800	23	1.618	0.872	0.255	4.225
zscore	37	15.478	9.389	1.374	51.132	23	14.842	8.680	4.685	39.884
iboone	37	0.019	0.197	−0.148	0.978	23	−0.066	0.077	−0.171	0.111

See Table 1 for the acronyms' description

Table 13 Descriptive statistics for the sample of European countries

Europe					
	count	mean	sd	min	max
coc	39	3.139	1.041	1.481	4.756
cpi	39	0.571	0.201	0.240	0.920
fl	39	0.399	0.148	0.140	0.710
bank	39	30.433	17.472	0.770	79.570
bancount	39	0.780	0.227	0.178	1.001
Save	39	0.322	0.225	0.010	0.784
credit	39	78.800	53.965	0.627	250.012
Borrowers	39	0.154	0.058	0.066	0.294
Deposit	39	0.867	0.102	0.517	0.987
Withdraw	39	0.886	0.082	0.617	0.990
atms	39	79.742	38.492	34.750	185.407
Debcard	39	0.646	0.258	0.157	0.986
Credcard	39	0.307	0.187	0.057	0.694
netpabuy	39	0.360	0.243	0.033	0.790
digpay	39	0.677	0.265	0.149	0.992
GDP	39	2.040	1.910	−1.145	7.849
npl	39	11.012	11.796	0.724	57.436
size	39	12.782	12.717	5.100	73.714
eficiency	39	57.528	13.223	29.346	90.967
regul	39	17.722	9.254	−1.935	62.242
roa	39	0.870	3.881	−5.918	21.800
zscore	39	12.057	7.666	2.724	39.205
iboone	39	−0.068	0.335	−1.319	0.654

See Table 1 for the acronyms' description

Table 14 Correlation matrix for the full sample of countries

Full sample

	zscore	lgroa	lgnpl	lgcoc	lgcpi	lgfl	ifi	innov	lgeficiency	lgregul	iboone	lgsize	lgrir	lgGDP	lginflat
lgzscore	1.00														
lgroa	0.00	1.00													
	(0.97)														
lgnpl	−0.25***	−0.01	1.00												
	(0.00)	(0.88)													
lgcoc	0.18**	−0.23***	−0.31***	1.00											
	(0.04)	(0.01)	(0.00)												
lgcpi	0.15*	−0.20**	−0.25***	0.91***	1.00										
	(0.08)	(0.02)	(0.00)	(0.00)											
lgfl	0.03	−0.06	−0.07	0.30***	0.29***	1.00									
	(0.70)	(0.51)	(0.43)	(0.00)	(0.00)										
ifi	0.01	−0.32***	−0.20**	0.47***	0.44***	0.20**	1.00								
	(0.92)	(0.00)	(0.02)	(0.00)	(0.00)	(0.02)									
innov	0.06	−0.28***	−0.32***	0.68***	0.61***	0.21**	0.43***	1.00							
	(0.52)	(0.00)	(0.00)	(0.00)	(0.00)	(0.01)	(0.00)								
lgeficiency	−0.02	−0.24***	−0.10	−0.02	−0.05	−0.07	0.07	−0.01	1.00						
	(0.84)	(0.01)	(0.24)	(0.83)	(0.57)	(0.45)	(0.41)	(0.91)							
lgregul	−0.07	0.10	−0.07	0.09	0.06	−0.14	−0.04	−0.07	0.15*	1.00					
	(0.39)	(0.26)	(0.39)	(0.30)	(0.48)	(0.11)	(0.61)	(0.39)	(0.07)						
iboone	−0.10	0.16*	0.19**	−0.08	−0.05	−0.02	−0.09	−0.05	−0.10	−0.27***	1.00				
	(0.23)	(0.07)	(0.02)	(0.33)	(0.57)	(0.80)	(0.32)	(0.56)	(0.25)	(0.00)					
lgsize	0.05	0.13	0.04	−0.21**	−0.18**	0.08	−0.08	−0.12	0.14*	−0.26***	0.15*	1.00			
	(0.57)	(0.14)	(0.67)	(0.02)	(0.04)	(0.38)	(0.36)	(0.15)	(0.10)	(0.00)	(0.07)				

Table 14 (continued)

Full sample

	zscore	lgroa	lgnpl	lgcoc	lgcpi	lgfl	ifi	innov	lgeficiency	lgregul	iboone	lgsize	lgrir	lgGDP	lginflat
lgrir	0.02	−0.11	0.17**	−0.20**	−0.17**	−0.07	−0.09	−0.20**	0.20**	0.02	0.07	0.13	1.00		
	(0.81)	(0.21)	(0.04)	(0.02)	(0.04)	(0.42)	(0.30)	(0.02)	(0.02)	(0.82)	(0.43)	(0.14)			
lgGDP	−0.00	0.01	−0.03	0.08	0.05	0.09	−0.03	−0.19**	−0.05	−0.03	−0.01	0.06	0.19**	1.00	
	(0.97)	(0.94)	(0.70)	(0.33)	(0.56)	(0.28)	(0.71)	(0.03)	(0.60)	(0.70)	(0.95)	(0.52)	(0.03)		
lginflat	−0.01	0.30***	−0.13	−0.39***	−0.37***	−0.11	−0.30***	−0.30***	−0.07	−0.00	−0.14*	−0.00	−0.04	−0.12	1.00
	(0.93)	(0.00)	(0.12)	(0.00)	(0.00)	(0.21)	(0.00)	(0.00)	(0.44)	(1.00)	(0.10)	(0.98)	(0.64)	(0.15)	

See Table 1 for acronyms' description. * $p<0.10$, ** $p<0.05$, and *** $p<0.01$

Author Contribution João Jungo was responsible for conceptualization, data curation, formal analysis, investigation, methodology, roles/writing–original draft. Mara Madaleno was responsible for investigation, methodology, project administration, resources, software, supervision, validation, visualization, roles/writing–original draft, and writing–review and editing. Anabela Botelho was responsible for resources, software, supervision, validation, visualization, roles/writing–original draft, and writing–review and editing. All authors have read and approved the final manuscript.

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Data Availability We declare that the data source are described in text. The data supporting the findings of this study are available from the first author upon reasonable request.

Declarations

Conflict of Interest The authors declare no conflict of interests.

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